

Course: B. Tech.

Branch: Civil Engineering

Semester: VIII

Subject Code & Name: BTCVSS801D & Maintenance & Repair of concrete Structure

Max Marks: 60

Date: 15-01-24

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) What are the reasons for failure of repairs?	CO1	6
B) Enlist different types of corrosions in concrete structures? Explain Carbonation induced corrosion in detail.	CO1	6
C) What is the necessity of corrosion management?	CO1	6
Q.2 Solve Any Two of the following.		12
A) Explain in brief Corrosion induced cracking and spalling in building.	CO2	6
B) State and explain different sources of sulphate which can attack on strength and durability of building structure.	CO2	6
C) What is DEF? Explain the mechanism of DEF induced cracking.	CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain in brief frost attack. Suggest suitable measures to reduce effect of frost attack.	CO3	6
B) Explain in detail with the help of suitable sketches, crack patterns developed due to ASR.	CO3	6
C) What are the different causes of deterioration of cementitious system?	CO3	6
Q.4 Solve Any Two of the following.		12
A) Explain the different steps of assessment of condition of RCC structures.	CO4	6
B) What is ground penetrating radar system? How it locate the delaminated concrete?	CO4	6
C) How the concrete surface repair is achieved?	CO4	6
Q. 5 Solve Any Two of the following.		12
A) State the different questions needs to be addressed during repair analysis?	CO5	6

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| B) | Explain the process of strengthening of slab and beams. | CO5 | 6 |
| C) | State the different methods of strengthening of column and explain any one in detail. | CO5 | 6 |

***** End *****